

Kino-VLA: Closed-Loop Embodied Reflection for Semantic Failure Recovery in Quadrupedal Locomotion

Anonymous Author(s)

Paper under double-blind review

Affiliation and contact details omitted for anonymity

Abstract—Learned controllers and rapid motor adaptation let legged robots reject a wide range of disturbances by reacting continuously to proprioceptive history. A distinct class of physical failures, however, cannot be recovered this way: escaping an elastic tether requires backing off rather than applying more torque, and a foot held by an adhesive trap is proprioceptively indistinguishable from one sinking into mud, yet the two demand opposite recoveries. Such failures require discrete, topological, strategy-level decisions that lie outside the hypothesis space of any continuous adaptive controller. We present Kino-VLA, a closed-loop embodied-reflection framework that couples high-frequency dynamics sensing with a vision-language recovery planner for a quadrupedal robot. Recovery is organized into five layers: a 1 kHz monitor-and-reflex loop that keeps the robot upright while reasoning proceeds; a Kino-Tokens extractor that distills 1 kHz proprioception into the language model’s embedding space under privileged-physics supervision; a vision-language planner that performs causal failure attribution, selects from an expanded recovery-primitive library, and maintains a persistent semantic traversability map; and a control-barrier-function safety shield that projects every proposed primitive onto a provably safe set, with a forward-invariance guarantee that, under a reduced-order model, the closed loop never leaves that set regardless of what the language model outputs. We further contribute Kino-Fail, a benchmark of eleven continuously parameterized failure operators with privileged ground truth and constructively matched attribution-ambiguity pairs, and a privileged-grounded hindsight chain-of-thought data pipeline whose truth-consistency filter rejects any annotation contradicted by simulator physics. In high-fidelity simulation on a physically simulated Unitree Go2, the trained planner attains 0.97 attribution accuracy on semantic-intervention failures, against 0.24 for a tuned rule-based state machine, while the safety shield solves in under 0.2 ms.

Index Terms—Legged robots, vision-language-action models, failure recovery, control barrier functions, embodied AI, safe learning, quadrupedal locomotion.

I. INTRODUCTION

Reinforcement learning has made quadrupedal locomotion remarkably robust. Policies trained in simulation with domain randomization, and augmented with online motor adaptation, cross rubble, mud, and ice and reject pushes that would topple a hand-tuned controller [1]–[4]. This robustness rests on continuous reaction: from a window of proprioceptive history the controller implicitly infers the local dynamics (friction, payload, terrain compliance) and adjusts its gait within a continuous parameter space [3]. For a broad family of disturbances this is the correct response, and we claim no improvement on it.

A separate family of failures defeats continuous reaction. A leg caught by an elastic tether is the canonical case: an adaptive controller raises torque to hold its commanded velocity, the tether stores that energy like a drawn bow, and the accumulated load eventually flings the robot off balance, so the safe action is the counter-intuitive one of backing off. Sudden resistance at a foot is a second case: if the cause is an adhesive trap the robot must step back, label the patch untraversable, and route around it, whereas if the cause is mud it must raise its gait and drive through. The recoveries are opposite, yet the two causes can be tuned to produce identical proprioceptive signatures, matched tangential-resistance curves, slip distributions, and torque spectra, so a controller reading proprioception alone cannot, even in principle, tell them apart.

We make this structure explicit as a *recoverability dichotomy*. A-class failures (uniform low friction, a static load offset, a mildly viscous floor) are recoverable by continuous adaptation, and an adaptive controller should handle them as well as any semantic system does. B-class failures require a discrete, topological, or strategy-level decision (back off, declare an entire region impassable, or abandon a torque-saturated task and request help), and an incorrect causal attribution produces the wrong, frequently opposite, recovery. What sets B-class failures apart is that the decision turns on a cause proprioception cannot resolve but vision can, such as the yellow sheen of a glue board against the brown of mud. These decisions lie outside the hypothesis space of any continuous adaptive controller and inside that of a semantic one. The scientific question we pose and answer empirically follows: which physical failures are recoverable by low-level adaptation, and which require semantic intervention?

Existing learning systems leave the B-class gap open. End-to-end vision-language-action models map observations to actions open-loop and at low frequency, without a force-feedback channel, and commit to a planned action through a developing failure [5]–[8]. Visually closed-loop planners react to RGB at camera rate [9], [10] and stay blind to a friction collapse or an actuator saturation until it has produced an overt visual symptom. Legged navigation models built on language backbones treat locomotion as a one-way black-box skill [11]–[13]: the planner issues velocity or waypoint commands but receives no upward report that the body is slipping or pinned. Closest to our aim, failure-reasoning systems such as RE-

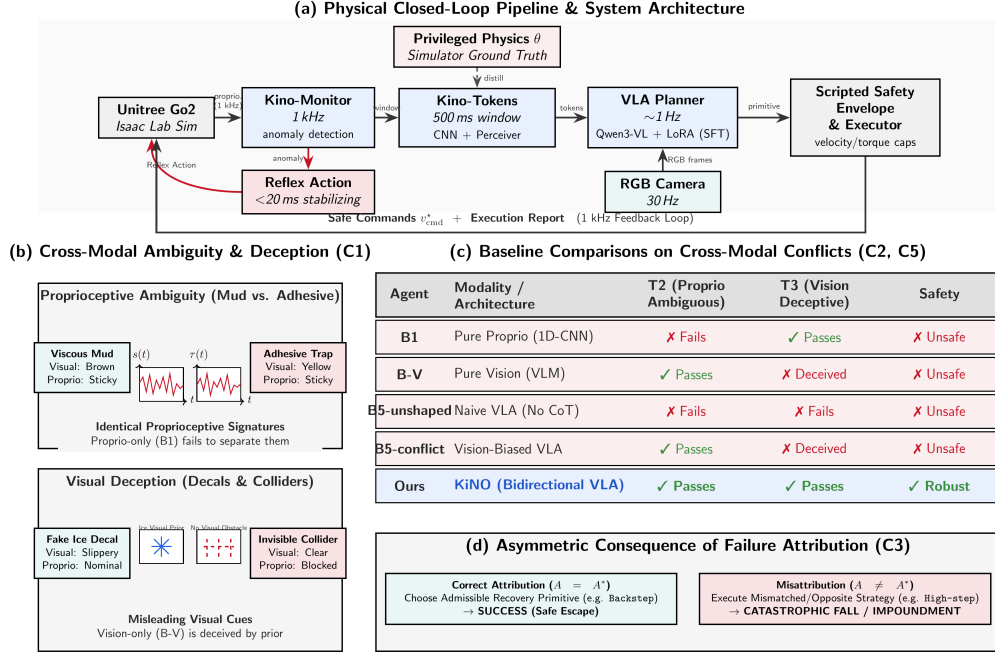


Fig. 1. **The Kino-VLA framework.** (a) **Recoverability dichotomy.** A-class failures (uniform low friction, a static load, a viscous floor) are recovered by continuous adaptation; B-class failures need a strategy-level decision. An adhesive trap (O4) and viscous mud (O2) produce *identical* proprioception—resistance, slip, and torque coincide—yet demand opposite recoveries (back off vs. switch gait), so only vision can decide. (b) **Kino-Fail benchmark.** Eleven parameterized failure operators on four mechanism axes, each exposing a privileged ground-truth parameter θ (friction, payload, effort, support); three constructively matched ambiguity pairs (O1↔O3, O2↔O4, O5↔O10) on which proprioception coincides; and five evaluation suites (Cal, Sem, Comp, Bound, OOD). (c) **Closed loop** on a physically simulated Unitree Go2. A 1 kHz Kino-Monitor and a millisecond reflex keep the robot upright; a Kino-Tokens extractor distills a 500 ms proprioceptive window into the planner under privileged-physics supervision; the ~ 1 Hz VLA planner—trained on the truth-filtered hindsight chain-of-thought data from (b)—attributes the failure, emits one recovery primitive, and reads and writes a persistent semantic traversability map, while the CBF-QP shield projects every primitive onto a provably safe set (<0.2 ms) before it is compiled to a velocity command.

FLECT [14] and AHA [15] summarize multimodal experience to explain and correct manipulation failures, but they attribute after the fact from low-frequency vision, address fixed-base manipulators rather than an underactuated floating-base robot, and inject conclusions as discrete text; on a legged platform they additionally lack the millisecond reflex that would keep the robot upright during the roughly one second of language-model inference.

We present Kino-VLA, a closed-loop embodied-reflection framework that supplies the missing upward channel from 1 kHz dynamics to a semantic planner and the missing downward guarantee that the planner cannot destabilize the robot. Kino-VLA organizes recovery into five layers (Fig. 1). A 1 kHz Kino-Monitor computes slip and tracking-error statistics and triggers a reflex loop that stabilizes the body (brake, stiffen, lower the center of mass) within milliseconds, bypassing the language model. A Kino-Tokens extractor compresses a 500 ms proprioceptive window with a 1D convolutional network and a Perceiver Resampler [16], [17] into fixed-length tokens projected into the language model’s embedding space; the latent is supervised by distillation against privileged simulator physics [3], [18], with an auxiliary physics-text

contrastive objective [19], so the tokens carry interpretable physical meaning and a calibrated out-of-distribution score. A roughly 1 Hz planner, a LoRA-adapted [20] vision-language model [21], [22], fuses vision, text, and Kino-Tokens to attribute the failure causally, selects a single primitive from an expanded recovery library, and maintains a persistent semantic traversability map onto which physical evidence is written. A control-barrier-function (CBF) safety shield and primitive compiler then project every proposed primitive onto a provably safe set before execution. Two ideas carry the design. First, *the language model proposes, the barrier function disposes*: a CBF quadratic program minimally edits each primitive so the capture point cannot leave a contracted support polygon, which yields a forward-invariance guarantee, under a reduced-order model, that the closed loop remains in its safe set whatever the language model outputs [23], [24]. Second, privileged physics grounds both the latent and the training data, replacing an unanchored black box with quantities the simulator can verify.

To study the dichotomy we introduce Kino-Fail, a benchmark of eleven continuously parameterized failure operators across four simulator mechanism axes, each exposing its parameter vector as privileged ground truth. Three operator

pairs are constructed so their proprioceptive statistics coincide, making a proprioception-only baseline fail on them by design, and a dedicated suite sweeps an operator’s parameter across the A/B frontier, turning the question of when semantic intervention becomes necessary into a measurable curve. To train the planner we build a privileged-grounded hindsight chain-of-thought [25] pipeline whose central component is an automated truth-consistency filter: an oracle annotation is kept only if its physical attribution matches the privileged parameters and its chosen primitive lies in the feasible recovery set, which converts “the oracle guessed” into “physically grounded.” The planner is then trained in two stages, supervised fine-tuning on the filtered data followed by an embodied preference-optimization stage [26] that scores rollouts by their physical outcome.

We implement and evaluate Kino-VLA in high-fidelity simulation on a physically simulated Unitree Go2 in Isaac Lab [27], [28]. On held-out semantic-intervention failures the trained planner attains an attribution accuracy of 0.97, against 0.24 for a tuned rule-based state machine, and every sampled output parses against the action schema; the safety shield solves its quadratic program in under 0.2 ms at the ninety-ninth percentile; and the privileged-distillation extractor regresses friction, payload, actuator effort, and support state to within per-operator tolerances. Sim-to-real transfer is treated at the level of design (an actuator-network model and contact-signal randomization) and left to future hardware studies.

This paper makes the following contributions.

- We formalize the *recoverability dichotomy* as a falsifiable question and instantiate it as Kino-Fail, a parameterized failure-operator benchmark with privileged ground truth and three constructively matched attribution-ambiguity pairs that render proprioception-only recovery provably insufficient.
- We cast the safety layer as a CBF quadratic program that arbitrates the unconstrained output of a language model, with a forward-invariance guarantee in a capture-point reduced-order model and an explicit dual-rate latency budget under which the reflex loop keeps the robot alive while the planner reasons.
- We ground the cross-frequency interface in privileged physics: a distilled Kino-Tokens latent with a calibrated out-of-distribution score, and a hindsight chain-of-thought pipeline whose truth-consistency filter rejects physically inconsistent annotations.
- We show, in simulation, that the resulting planner attributes and recovers from semantic-intervention failures where a tuned rule-based baseline cannot, evidence for the irreplaceability of vision-grounded reasoning on the B-class subspace.

II. RELATED WORK

A. Learned legged locomotion and online adaptation

Model-free reinforcement learning produces quadrupedal controllers that walk over challenging terrain [1], [2] and adapt

online to unmodeled dynamics by encoding proprioceptive history into a latent that drives the policy [3], [4]. These controllers are trained at scale with massively parallel simulation [27]–[29] and transferred to hardware with actuator-network models of motor and gearbox nonlinearity [30]. Most use teacher-student privileged distillation, in which a student trained on onboard observations regresses a teacher’s access to privileged simulator state [1], [18]; we reuse exactly this mechanism for Kino-Tokens. The boundary is deliberate. These methods encode dynamics to drive a continuous control policy, and we make no claim to improve their adaptation. We distill the same privileged signal into a *semantic* embedding that supports discrete decisions, and we target the B-class subspace where continuous adaptation is insufficient by construction rather than merely suboptimal.

B. Vision-language-action models and embodied foundation models

Large multimodal models have been adapted into robot policies that map language and pixels to actions [5]–[8], [31]–[33], and into high-level reasoners that sequence skills through language [34]–[36]. Legged variants extend this recipe to navigation [11]–[13], and visually closed-loop systems re-plan from RGB observations as a task unfolds [9], [10]. In all of these, action is generated open-loop or closed only on vision at camera rate, and on legged platforms locomotion is invoked as a black-box skill that returns no dynamics feedback. Kino-VLA adds a 1 kHz proprioceptive channel into the planner and a barrier-certified compiler beneath it, so the planner both sees the failure in the dynamics and cannot issue an action that destabilizes the body.

C. Failure detection, attribution, and language-model recovery

A growing line of work uses language and vision models to explain and correct robot failures. REFLECT [14] summarizes multimodal robot experience into a hierarchical trace for failure explanation, AHA [15] trains a vision-language model to detect and reason over manipulation failures, and DoReMi [37] detects and recovers from plan-execution misalignment. In parallel, classical legged-robot pipelines detect slip and anomalies from proprioception [38], and introspective perception predicts when a vision system will fail [39]. The distinctions are several. Attribution in the language-model line is performed after the fact, from low-frequency vision or audio, on fixed-base manipulators, and is injected as discrete text; classical legged anomaly detection stops at raising an alarm. Kino-VLA attributes *online*, during a reflex-maintained safe interval, from a 1 kHz dynamics channel, and closes the loop through an expanded recovery-primitive library and a persistent map rather than terminating at an explanation or an alarm.

D. Safe control and safety filters

Control barrier functions provide quadratic-program safety filters with forward-invariance guarantees [23], [40], and

TABLE I
END-TO-END LATENCY BUDGET OF THE DUAL-RATE LOOP.

Stage	Budget
Anomaly $\rightarrow$ Kino-Monitor detection	< 5 ms
Reflex stabilization	< 20 ms
Kino-Token window fill	500 ms
VLA inference	~ 1 s
CBF arbitration + primitive compile	< 10 ms

legged balance has long been analyzed through reduced-order capture-point [24] and divergent-component-of-motion [41] models built on the linear inverted pendulum and zero-moment-point formulation [42]. Recent work layers barrier functions with model predictive control for legged robots [43], certifies learning-based controllers with predictive safety filters [44], and surveys the broader safe-learning program [45]. These methods certify a *known* controller against bounded disturbances. We instead place the CBF quadratic program downstream of an unconstrained, potentially hallucinated language-model output and project discrete recovery primitives onto a capture-point safe set, which addresses formal safety for language-model-in-the-loop legged control.

E. Semantic maps and open-vocabulary perception for navigation

Open-vocabulary scene representations build navigable maps by grounding vision into 3D with CLIP [46] and SAM [47] features [48]–[50], and self-supervised traversability estimation learns terrain costs from experience [51]. These approaches write traversability from vision (vision $\rightarrow$ map). Kino-VLA inverts the direction: a physical failure rewrites the visually derived label of its 3D region and propagates the correction across visually homogeneous neighbors (physics $\rightarrow$ map), so that stepping through a single broken patch of thin ice condemns the entire visually uniform sheet.

III. THE KINO-VLA SYSTEM

A. Overview

Kino-VLA closes the loop from a 1 kHz dynamics stream to a roughly 1 Hz semantic planner and back, in five layers running at three rates. A *Kino-Monitor* reads proprioception at 1 kHz and raises an anomaly flag; a *reflex loop* stabilizes the body within milliseconds; a *Kino-Tokens extractor* compresses a 500 ms dynamics window into the planner’s embedding space; a *VLA recovery planner* attributes the failure, selects a recovery primitive, and updates a semantic map; and a *CBF safety shield* with a primitive compiler runs at 500 Hz–1 kHz. The dual-rate design rests on the latency budget in Table I: the reflex holds the state inside the safe set for a dwell time that covers the planner’s inference latency, while the shield guarantees that whatever the planner eventually outputs cannot push the state out of the safe set. The reflex and the shield together make the system safe in the A-class regime; the planner supplies the semantic decision the B-class regime requires.

B. Kino-Tokens: privileged-physics distillation

The Kino-Monitor computes, at 1 kHz, a slip score and a velocity-tracking error from onboard proprioception (joint torques estimated from motor current, joint angles, IMU attitude). Crossing calibrated thresholds arms the reflex loop, which applies a damping stand, a wider stance, and a lower center of mass without invoking the planner, and gates the extractor on.

A 500 ms sliding window of proprioception is encoded by a 1D convolutional network, which captures local transients, followed by a Perceiver Resampler [16], [17] that cross-attends the window into fixed-length tokens z . The latent is supervised primarily by distillation against privileged simulator physics, in the teacher-student style of privileged learning [3], [18]: a regression head g predicts the privileged parameter vector θ (true friction μ , contact stiffness, payload mass, actuator effort limit) under a Huber loss $\mathcal{L}_\theta = \text{Huber}(g(z), \theta)$, with an auxiliary InfoNCE [19] objective aligning z to text. Privileged supervision buys three properties: the latent is physically interpretable, out-of-distribution detection becomes the *measurable* residual of the θ prediction rather than an abstract notion of manifold departure, and the same residual grounds the data filter of Sec. III-E. A Kino-Projector multilayer perceptron maps z into the vision-language model’s embedding space as soft tokens, and anomaly-gated injection zeroes their weight at steady state, waking the channel only on an alarm. The online estimate $\hat{\mu}$ is forwarded to the shield, coupling perception to safety.

C. Recovery planner and semantic traversability map

The planner is a pretrained vision-language model (Qwen3-VL-4B [21], the successor to Qwen2-VL [22]) adapted with LoRA [20]. At each step it consumes recent RGB frames, a textual context (task, history, a crop of the current map, and any rejection codes from the shield), and the Kino-Tokens as soft tokens; a text route that serializes the physics summary into words replaces the soft tokens as an ablation arm. The planner emits a structured response, a `<Thought>` chain of reasoning [25] followed by a `<Action>` that names one primitive and its parameters. A schema-validating parser admits only well-formed atomic actions, so no malformed command reaches the compiler.

The primitive library spans the locomotion policy’s command space: `Backstep`, `Replan_Waypoint` (a 2D pixel goal back-projected to a 3D waypoint), `Switch_Gait`, `Adjust_Posture`, `Set_Constraint`, `Update_Topology`, and `Hold_and_Request`. The attribution-to-recovery mapping is many-to-many by design: identical proprioception can demand opposite primitives, so a correct recovery depends on a vision-grounded causal attribution that a controller over proprioception cannot make.

The semantic traversability map persists topological memory across viewpoints. Open-vocabulary segmentation with CLIP [46] and SAM [47] features partitions RGB into regions, and RGB-D depth back-projection lifts them to 3D

regions in the odometry frame, in the spirit of language-grounded maps [48], [50] but driven in the opposite direction. A visual prior seeds traversability; when a physical failure occurs, the Kino-Tokens attribution overwrites the corresponding region’s costmap label as high-confidence evidence, and CLIP-similarity propagation extends the correction to visually homogeneous neighbors, so a single broken patch of thin ice condemns the whole sheet. The map crop returned to the planner each round removes the “forget on turning around” failure of prompt-only memory.

D. CBF-QP safety shield

The shield treats the planner as an untrusted proposer and a control barrier function as the arbiter. We adopt a reduced-order linear inverted pendulum for the center of mass at constant height z_c , set by the posture mode σ :

$$\dot{p} = v, \quad \dot{v} = \omega^2(p - u), \quad \omega = \sqrt{g/z_c}, \quad (1)$$

where $u \in \mathbb{R}^2$ is the zero-moment point that the whole-body controller realizes through foot forces. The divergent component of motion [41] (the capture point [24]) $\xi = p + v/\omega$ obeys $\dot{\xi} = \omega(\xi - u)$; because the complementary dynamics $\dot{p} = \omega(\xi - p)$ are exponentially stable, constraining ξ constrains the only unstable mode. Writing the contracted support polygon as $\{y : a_j^\top y \leq b_j, \|a_j\| = 1\}$ with margin $\delta > 0$, we define per-edge barriers and the safe set

$$h_j(x) = (b_j - \delta) - a_j^\top \xi, \quad \mathcal{C} = \{x : h_j(x) \geq 0 \ \forall j\}, \quad (2)$$

which encodes 0-step capturability: while ξ lies inside the contracted polygon a no-step ZMP can bring the robot to rest. Differentiating h_j and imposing the barrier condition $\dot{h}_j \geq -\alpha h_j$ gives a constraint linear in u ,

$$\omega a_j^\top u \geq \omega a_j^\top \xi - \alpha h_j(x). \quad (3)$$

The compiler maps a primitive to a desired velocity v_{cmd} and mode σ ; with $\xi_{\text{des}} = p + v_{\text{cmd}}/\omega$ a DCM-tracking law yields the nominal input $u_{\text{nom}} = \xi + K_\xi(\xi - \xi_{\text{des}})$. Every control cycle the shield solves the convex program

$$\begin{aligned} u^* &= \arg \min_{u \in \mathbb{R}^2} \frac{1}{2} \|u - u_{\text{nom}}\|^2 \\ \text{s.t. } &\omega a_j^\top u \geq \omega a_j^\top \xi - \alpha h_j(x), \quad \forall j, \\ &A_S u \leq b_S - \delta_u \mathbf{1}, \\ &\|p - u\| \leq \hat{\mu} z_c, \end{aligned} \quad (4)$$

whose three constraint blocks are the CBF condition (3), ZMP realizability under unilateral contact, and the friction cone with the online estimate $\hat{\mu}$; the issued command is back-solved as $v_{\text{cmd}}^* = v + (\omega/K_\xi)(\xi - u^*)$. The program has two variables and $O(M)$ linear constraints and is solved by exact projection.

Forward invariance. If $x(0) \in \mathcal{C}$ and (3) holds each cycle, then $h_j(t) \geq h_j(0)e^{-\alpha t} \geq 0$, so $x(t) \in \mathcal{C}$ for all t under the reduced-order model: the capture point never leaves the contracted support polygon, whatever the planner outputs [23], [40]. Discrete mode switches change $\mathcal{C}$ itself, so a switch $\sigma \rightarrow \sigma'$ is admitted only when $h_j^{\sigma'}(x) \geq \varepsilon_{\text{switch}} \forall j$, and is otherwise rejected with a structured code returned to the

planner. If the QP is infeasible, the shield does not relax the safety constraints; it falls back to a reflex damping stand that widens the stance to enlarge the support polygon. Because the friction bound carries $\hat{\mu}$, detecting ice automatically contracts the admissible ZMP set and renders the shield conservative, the explicit coupling between the perception and safety layers.

E. Training: hindsight CoT, truth-consistency filter, SFT and DPO

Training data come from a four-phase privileged-grounded hindsight pipeline. A procedural counterfactual maze randomizes the binding between appearance and physics, forcing the model to reconstruct traversability from the conflict between Kino-Tokens and RGB rather than from visual priors. The Kino-Monitor intercepts each failure and packages a multi-modal snapshot: five RGB and depth frames, the Kino-Tokens window, the prior planner outputs, and the privileged physics θ . A frontier vision-language model, shown the snapshot but not the privileged class label, writes a hindsight chain of thought ending in a primitive choice. The core component is an automated *truth-consistency filter*: a sample is kept only if the narrative’s physical attribution matches θ and its chosen primitive lies in the failure’s feasible recovery set, and is discarded whole otherwise. Because the ambiguity pairs are constructed to be disjoint on their discriminating primitive, a fluent “right story, wrong-sibling strategy” annotation is droppable; the filter converts a possibly confabulated oracle narrative into physically grounded supervision.

Kino-SFT then fine-tunes the planner with LoRA on the filtered data. A second stage of embodied preference optimization [26] places the model in the closed loop, samples rollouts at each failure node, and forms preference pairs from the physical outcome (escape and reach versus fall or deadlock); by the disjoint-primitive construction, a wrong-sibling strategy at an ambiguity node is automatically a high-quality rejected sample, so the objective optimizes attribution correctness directly.

IV. EXPERIMENTS

A. Setup

Platform. All experiments run in Isaac Lab [27], [28] on a physically simulated Unitree Go2. The low-level velocity-tracking policy is trained with massively parallel reinforcement learning [29] under friction and mass domain randomization with an anti-skating reward, and motor nonlinearity follows an actuator-network model [30]. We state plainly that every result below is in high-fidelity simulation; there is no hardware experiment, and sim-to-real transfer is addressed only at the level of design.

Benchmark. Kino-Fail comprises eleven failure operators on four mechanism axes (contact material field, external wrench and attachment, geometry and visual-physics decoupling, and embodiment degradation), each carrying a continuous privileged parameter θ . Five suites organize them: Cal (A-class calibration), Sem (B-class semantic), Comp (operator superposition, test only), Bound (a θ -sweep across the A/B fron-

TABLE II

ATTRIBUTION ACCURACY ON HELD-OUT SUITE-SEM AMBIGUITY PAIRS ($n = 38$). PARSE IS THE ACTION-SCHEMA PARSE RATE; FEAS. IS THE FEASIBLE-RECOVERY RATE; N/A MARKS METRICS NOT DEFINED FOR THE RULE MACHINE.

Method	Attribution	Parse	Feas.
Rule-based FSM (B2)	0.237	n/a	n/a
Kino-VLA-text (B4)	0.974	1.000	0.974
Kino-VLA-latent (B5)	0.974	1.000	0.974

tier), and OOD (leave-one-operator-out plus parameter extrapolation). Suite-Sem centers on three constructively matched attribution-ambiguity pairs whose proprioceptive statistics coincide: adhesion versus viscous mud, thin-ice collapse versus uniform ice, and payload overload versus actuator decay. In each pair vision must decide the recovery.

Model and data. The planner is Qwen3-VL-4B with LoRA (34.5 M trainable parameters). It is trained on a 413-sample hindsight chain-of-thought dataset collected on the simulated Go2 and annotated by a frontier vision-language oracle, then truth-filtered; the operator-stratified split is 315/49/49.

Baselines. Following the evaluation design we consider B1 (an adaptive, no-LLM controller), B2 (a tuned rule-based finite-state machine, no LLM), B3 (a vision-reflection text-only planner), B4 (Kino-VLA with the text route), and B5 (Kino-VLA with the latent route, the full system). The comparisons reported here center on B5 and B4 against the rule-based FSM (B2) and on the latent-versus-text ablation; the complete five-baseline sweep across all suites, including the Suite-Cal check that an adaptive baseline matches B5 on A-class failures, is ongoing.

B. Attribution on semantic-intervention failures

Table II reports attribution accuracy on the held-out Suite-Sem ambiguity pairs ($n = 38$). Kino-VLA reaches 0.974 against 0.237 for the tuned rule-based FSM, a margin of +0.737, with a perfect action-schema parse rate and a 0.974 feasible-recovery rate; per operator the accuracy is 0.89 on O1 and 1.00 on O2, O3, O4, O5, and O10. Lacking a vision-grounded attribution, the FSM collapses each ambiguity pair to a single proprioceptive response and is correct only when that response happens to match. This is the B-class irreplaceability result: where the correct recovery hinges on a cause proprioception cannot resolve, a rule machine over proprioception cannot succeed, and no tuning of B2 closes the gap.

Latent versus text ablation. The text route (B4) also reaches 0.974 attribution. On these snapshots vision together with either proprioceptive channel suffices for attribution, so the latent channel’s measured benefit is finer recovery-parameter precision (for instance the backstep distance or the speed cap), not attribution accuracy. We report this rather than claim a latent advantage the data do not support.

TABLE III

PRIVILEGED-PHYSICS REGRESSION ERROR (MEAN ABSOLUTE ERROR) PER CHANNEL, ON THE CPU SURROGATE AND THE SIMULATED Go2, AGAINST THE PER-OPERATOR TOLERANCE. ALL CHANNELS PASS ON BOTH BACKENDS.

Channel	Surrogate	Simulated Go2	Tol.
Friction μ	0.027	0.071	0.10
Payload m (kg)	1.35	0.97	1.50
Actuator effort	0.055	0.069	0.10
Support ratio	0.013	0.092	0.12

C. Privileged-physics distillation

The Kino-Tokens extractor regresses θ to within per-operator tolerances both on the CPU surrogate and on the simulated Go2 (Table III). Each channel is scored on its observable regime; we report, rather than mask, a genuine observability floor where mid-range friction and small payloads are compensated by healthy actuators and leave no proprioceptive trace. The out-of-distribution residual rises monotonically under parameter extrapolation (Spearman correlation 1.0, separation 1.88), and inference runs at 0.6–0.8 ms at the ninety-ninth percentile against a 10 ms budget. Feeding the online estimate $\hat{\mu}$ to the shield contracts its friction radius from 0.247 m to 0.057 m on detected ice, the perception-to-safety coupling of Sec. III-D in action.

D. CBF-QP safety shield

The shield solves (4) by exact projection in 0.11–0.16 ms at the ninety-ninth percentile, comfortably inside the 1 ms budget. Under a stream of hostile velocity commands it intervenes on every out-of-set command, clamping the issued speed from 2.50 m/s to at most 1.99 m/s, and on the surrogate adversarial protocol it produces zero falls while the bypassed controller falls. The reflex loop extends quasi-static survival from 2.1 s to 20 s, supplying the safe dwell that the latency budget (Table I) requires.

We report one finding without varnish. On the full-order learned policy, a capture-point push sweep ($4J \times$ three directions $\times$ three seeds, 36 scenarios) shows that the reduced-order CBF does *not* reduce falls and is anti-protective for forward and diagonal pushes, where its braking destabilizes a policy that would otherwise ride the push out. The zero-fall guarantee is therefore a property of the reduced-order model of Sec. III-D, and the demonstrable claim on the full-order robot is command clamping together with the formal forward-invariance result; closing the gap between the reduced-order certificate and the full-order policy is left to future work.

E. Data pipeline and the truth-consistency filter

The truth-consistency filter is golden-tested against synthetic confabulated chains of thought spanning all of its verdict codes, and on real oracle output it catches genuine vision-language-model confabulations rather than only synthetic ones. On a 200-cell surrogate build it keeps 147 samples (26.5% rejected) at 7.3×10^5 samples per hour; on the simulated-Go2 dataset the kept set is 413 at roughly 21% reject

once oracle and transport errors are separated from genuine filter rejections. The dataset card reports an A/B balance of 199/214 and all three ambiguity pairs present on both sides. The set is proof-of-pipeline scale, which the card flags.

F. Embodied preference optimization

On 242 in-distribution ambiguity-sibling preference pairs the DPO objective raises preference accuracy from 0.88 to 1.00: the model perfectly prefers the correct recovery over the wrong-sibling strategy, which verifies the intended mechanism. The strong SFT, however, already saturates held-out top-one attribution (0.974, 0.912, and 0.882 at decoding temperatures 0, 0.8, and 1.2), so the preference-optimized model ties SFT without regression but without top-one headroom, and an on-policy variant trained on the hardest confidently-wrong cases was undertrained at our configuration. We therefore report the preference mechanism as verified, but we do not claim a closed-loop success margin of DPO over SFT; establishing one, with a tuned on-policy stage at scale and a finer recovery-parameter metric, is future work.

G. Closed-loop demonstration and limitations

End to end, the trained planner runs inside the Isaac closed loop: the monitor fires, the planner attributes from the live RGB and proprioception, the parser validates the action, and the CBF compiler executes it on the simulated Go2. Over three Suite-Sem scenarios the SFT planner reaches the goal once under greedy decoding. We explain rather than hide the shortfall. The hindsight snapshots are captured under bang-bang excitation, which is needed so that payload and actuator decay are observable for labeling, whereas the deployed planner navigates smoothly, so the runtime proprioception is out of distribution; and the invisible-collider operator is absent from the training set, so it defaults to a wrong primitive. The clean quantitative evidence is therefore the in-distribution attribution of Table II, and the closed loop is the full-stack demonstration. Bridging the train-deploy proprioception shift, scaling the dataset past proof-of-pipeline, and running the complete five-baseline sweep across all five suites are the priorities of ongoing work.

V. CONCLUSION

We have argued, and shown in simulation, that not every physical failure of a legged robot is recoverable by continuous adaptation, and that the failures which are not, the B-class failures, turn on a cause proprioception cannot resolve but vision can. Kino-VLA closes the missing loop around this subspace: a 1 kHz monitor and reflex keep the robot alive, a privileged-distilled Kino-Tokens latent carries the dynamics into a vision-language planner, the planner attributes the failure and selects a recovery primitive while writing a persistent semantic map, and a control-barrier-function shield arbitrates every primitive so that, under a reduced-order model, the closed loop cannot leave its safe set. On the Kino-Fail benchmark, whose constructive ambiguity pairs make proprioception-only recovery insufficient by design, the trained planner attributes B-class

failures at 0.974 against 0.237 for a tuned rule-based machine, evidence that vision-grounded reasoning is irreplaceable where adaptation alone is not enough.

The work is bounded in ways we have made explicit and that set the agenda. All results are in high-fidelity simulation; transferring the high-frequency channel to hardware, through an actuator-network model and contact-signal randomization, is the next step. The shield’s zero-fall guarantee holds in the reduced-order model but not on the full-order policy, where its demonstrated role is command clamping. Narrowing that gap, scaling the hindsight dataset past proof-of-pipeline, demonstrating a closed-loop margin for embodied preference optimization, and running the complete five-baseline sweep across all five suites are the priorities ahead.

REFERENCES

- [1] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning quadrupedal locomotion over challenging terrain,” *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.
- [2] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning robust perceptive locomotion for quadrupedal robots in the wild,” *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.
- [3] A. Kumar, Z. Fu, D. Pathak, and J. Malik, “RMA: Rapid motor adaptation for legged robots,” in *Robotics: Science and Systems (RSS)*, 2021.
- [4] I. M. A. Nahrendra, B. Yu, and H. Myung, “DreamWaQ: Learning robust quadrupedal locomotion with implicit terrain imagination via deep reinforcement learning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [5] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choremanski, T. Ding, D. Driess *et al.*, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” in *Conference on Robot Learning (CoRL)*, 2023.
- [6] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, “Open-VLA: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [7] Octo Model Team, D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu *et al.*, “Octo: An open-source generalist robot policy,” in *Robotics: Science and Systems (RSS)*, 2024.
- [8] D. Driess, F. Xia, M. S. M. Sajjadi, C. Lynch, A. Chowdhery, B. Ichter, A. Wahid, J. Tompson, Q. Vuong, T. Yu *et al.*, “PaLM-E: An embodied multimodal language model,” in *International Conference on Machine Learning (ICML)*, 2023.
- [9] Q. Bu, J. Zeng, L. Chen, Y. Yang, G. Zhou, J. Yan, P. Luo, H. Cui, Y. Ma, and H. Li, “Closed-loop visuomotor control with generative expectation for robotic manipulation,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [10] P. Zhi, Z. Zhang, Y. Zhao, M. Han, Z. Zhang, Z. Li, Z. Jiao, B. Jia, and S. Huang, “Closed-loop open-vocabulary mobile manipulation with GPT-4V,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [11] A.-C. Cheng, Y. Ji, Z. Yang, Z. Gongye, X. Zou, J. Kautz, E. Bıyık, H. Yin, S. Liu, and X. Wang, “NaVILA: Legged robot vision-language-action model for navigation,” *arXiv preprint arXiv:2412.04453*, 2024.
- [12] Y. Mei, Y. Wang, S. Zheng, and Q. Jin, “QuadrupedGPT: Towards a versatile quadruped agent in open-ended worlds,” *arXiv preprint arXiv:2406.16578*, 2024.
- [13] Q. Wu, Z. Fu, X. Cheng, X. Wang, and C. Finn, “Helpful DoggyBot: Open-world object fetching using legged robots and vision-language models,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [14] Z. Liu, A. Bahety, and S. Song, “REFLECT: Summarizing robot experiences for failure explanation and correction,” in *Conference on Robot Learning (CoRL)*, 2023.

- [15] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," in *International Conference on Learning Representations (ICLR)*, 2025.
- [16] A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, and J. Carreira, "Perceiver: General perception with iterative attention," in *International Conference on Machine Learning (ICML)*, 2021.
- [17] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds *et al.*, "Flamingo: a visual language model for few-shot learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [18] D. Chen, B. Zhou, V. Koltun, and P. Krähenbühl, "Learning by cheating," in *Conference on Robot Learning (CoRL)*, 2020.
- [19] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *arXiv preprint arXiv:1807.03748*, 2018.
- [20] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations (ICLR)*, 2022.
- [21] S. Bai, Y. Cai, R. Chen *et al.*, "Qwen3-v1 technical report," *arXiv preprint arXiv:2511.21631*, 2025.
- [22] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, and J. Lin, "Qwen2-v1: Enhancing vision-language model's perception of the world at any resolution," *arXiv preprint arXiv:2409.12191*, 2024.
- [23] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017.
- [24] J. Pratt, J. Carff, S. Drakunov, and A. Goswami, "Capture point: A step toward humanoid push recovery," in *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2006, pp. 200–207.
- [25] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [26] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [27] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac gym: High performance GPU-based physics simulation for robot learning," *arXiv preprint arXiv:2108.10470*, 2021.
- [28] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar, A. Mandlekar, B. Babich, G. State, M. Hutter, and A. Garg, "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3740–3747, 2023.
- [29] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, "Learning to walk in minutes using massively parallel deep reinforcement learning," in *Conference on Robot Learning (CoRL)*, 2022.
- [30] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019.
- [31] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog *et al.*, "RT-1: Robotics transformer for real-world control at scale," in *Robotics: Science and Systems (RSS)*, 2023.
- [32] X. Li, M. Liu, H. Zhang, C. Yu, J. Xu, H. Wu, C. Cheang, Y. Jing, W. Zhang, H. Liu, H. Li, and T. Kong, "Vision-language foundation models as effective robot imitators," in *International Conference on Learning Representations (ICLR)*, 2024.
- [33] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter *et al.*, " π_0 : A vision-language-action flow model for general robot control," 2024, physical Intelligence technical report.
- [34] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, C. Fu, K. Gopalakrishnan, K. Hausman *et al.*, "Do as i can, not as i say: Grounding language in robotic affordances," in *Conference on Robot Learning (CoRL)*, 2022.
- [35] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng, "Code as policies: Language model programs for embodied control," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 9493–9500.
- [36] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar, P. Sermanet, N. Brown, T. Jackson, L. Luu, S. Levine, K. Hausman, and B. Ichter, "Inner monologue: Embodied reasoning through planning with language models," in *Conference on Robot Learning (CoRL)*, 2022.
- [37] Y. Guo, Y.-J. Wang, L. Zha, Z. Jiang, and J. Chen, "DoReMi: Grounding language model by detecting and recovering from plan-execution misalignment," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
- [38] C. Yan, J. Qin, Q. Liu, and Q. Ma, "Slip detection and recovery for quadruped robots via orthogonal decomposition," *IEEE Transactions on Industrial Electronics*, vol. 72, no. 6, pp. 6166–6174, 2025.
- [39] S. Daffry, S. Zeng, J. A. Bagnell, and M. Hebert, "Introspective perception: Learning to predict failures in vision systems," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016.
- [40] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *European Control Conference (ECC)*, 2019, pp. 3420–3431.
- [41] J. Engelsberger, C. Ott, and A. Albu-Schäffer, "Three-dimensional bipedal walking control based on divergent component of motion," *IEEE Transactions on Robotics*, vol. 31, no. 2, pp. 355–368, 2015.
- [42] S. Kajita, F. Kanehiro, K. Kaneko, K. Fujiwara, K. Harada, K. Yokoi, and H. Hirukawa, "Biped walking pattern generation by using preview control of zero-moment point," in *IEEE International Conference on Robotics and Automation (ICRA)*, vol. 2, 2003, pp. 1620–1626.
- [43] R. Grandia, A. J. Taylor, A. D. Ames, and M. Hutter, "Multi-layered safety for legged robots via control barrier functions and model predictive control," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 8352–8358.
- [44] K. P. Wabersich and M. N. Zeilinger, "Linear model predictive safety certification for learning-based control," in *IEEE Conference on Decision and Control (CDC)*, 2018, pp. 7130–7135.
- [45] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 5, pp. 411–444, 2022.
- [46] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021.
- [47] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, "Segment anything," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [48] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [49] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "Voxposer: Composable 3d value maps for robotic manipulation with language models," in *Conference on Robot Learning (CoRL)*, 2023.
- [50] B. Chen, F. Xia, B. Ichter, K. Rao, K. Gopalakrishnan, M. S. Ryoo, A. Stone, and D. Kappler, "Open-vocabulary queryable scene representations for real world planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [51] J. Frey, M. Mattamala, N. Chebrolu, C. Cadena, M. Fallon, and M. Hutter, "Fast traversability estimation for wild visual navigation," in *Robotics: Science and Systems (RSS)*, 2023.